

P-VM-005 — Tram TV-100 fleet — repair, overhaul or replace — option study

Option	Capex (EUR)	Life gained	Downtime per tram	Assumptions
Repair on condition	€9.8M	5 years	12 days	No structural findings
Mid-life overhaul (selected)	€29.9M	15 years	45 days	Bogie inspection only, no frame replacement
Replace (24 new trams)	€96M	30 years	—	Depot adaptation extra

Key assumption of the selected option: bogie frames need inspection, not replacement. If NDT finds frame cracking on more than 10% of frames, the option must be re-appraised.

Lifecycle cost comparison (15 years, constant 2023 prices)

Option	Capex	Maintenance (15 y)	Availability
Repair on condition	€9.8M	€41.0M	91%
Mid-life overhaul	€29.9M	€28.4M	96%
Replace	€96.0M	€18.2M	98%

Discount rate 4%, residual value excluded, funding perspective: the Authority (PPI). The operator's perspective differs: maintenance is inside the contract price until 2028.